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DC014 COMPUTER SCIENCE
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NAME: _____
MATRIC NO: _____
PRACTICUM: _____

Answer all questions.

- 1** Adam is a content creator for ABC company. He always makes live broadcasts to promote a new product to his followers.

- (a)** Name **two (2)** most suitable multimedia elements that Adam can use to promote his product on social media platforms.

[2 marks]

- (b)** Describe **one (1)** purpose of using the multimedia element mentioned in question **1(a)**.

[2 marks]

- (c)** Identify **two (2)** common file formats mentioned in **1(b)** that Adam used on his social media platform.

[2 marks]

- 2** A multimedia product is a product that combines different content forms such as text, audio, images, animation, or video into one single product. Multimedia products can be used for a variety of purposes, including education, entertainment, and marketing.

- (a)** List **three (3)** processes that need to go through before a multimedia application is produced.

[3 marks]

- (b)** State the process involved based on the statement given below:

[4 marks]

Statement	Process
The first question we ask is “why” do you want to develop a multimedia project?	
The edited scripts are used to plan the location, performers, time schedules and budget, then the shoot is scheduled.	
The product is tested on multiple computers and monitors.	
Determine the basic of what types of equipment the audience have and what obstacles must be overcome.	

- (c)** Authoring tools enable users to create a digital course and publish it in select formats. There are **three (3)** types of authoring tools. Identify each type.

[3 marks]

- (d) Describe **two (2)** differences between these types of authoring tools. [2 marks]

- (e) HD media player or HDD media player (HDMP) is a consumer product that combines a digital media player with a hard drive (HD) enclosure with all the hardware and software for playing audio, video, and photos on a television.

- i) HD media player is one example of offline multimedia delivery. Give **two (2)** examples of online multimedia delivery which can also work as an HD media player.

[2 marks]

- ii) Describe **one (1)** difference between offline and online multimedia delivery.

[2 marks]

- 3 Many actions we do every day like using an ATM, shopping online and many more generate data that needs to be stored and managed and used by others.

- (a) Define database.

[2 marks]

- (b) Give **one (1)** difference between a file processing system and a database system.

[2 marks]

File Processing System	Database System

- (c) Describe **one (1)** advantage of using a database system.

[2 marks]

- 4 Database can manage large amounts of data efficiently and enable information to be shared.

DeptCode	DeptName	NoOfLecturers
D01	Chemistry	32
D02	Computer Science	28
D03	Biology	29

- (a) Based on the table above, state the most suitable primary key.

[1 mark]

- (b) State the data type for **DeptName**.

[1 mark]

- (c) Describe **one (1)** function of a data dictionary.

[2 marks]

- 5 One of the major reasons businesses, homes and others purchase computers is for Internet access. The Internet is a widely used research tool, providing society with access to global information and instant communications.

- (a) Explain how the Global Positioning System (GPS) works.

[2 marks]

- (b) State **two (2)** other uses of the network besides the one mentioned in 5(a).

[2 marks]

- (c) State the type of wired transmission media based on usage given.

[4 marks]

Usage	Type of wired transmission media
Better security for signals during transmission because they are less susceptible to noise.	
One of the more widely used transmission media for network cabling and telephone systems.	
Primarily used by cable TV companies to connect their satellite antenna facilities to customer homes and businesses.	

This type of transmission media is faster, more reliable and more expensive.	
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- (d) State the type of wireless transmission media based on given characteristics.

[4 marks]

Characteristic	Type of wireless transmission media
Enables the transfer of data from one device to another device such as a mobile computer, mouse, printer and smartphone.	
The type of transmission media in the form of broadcast radio is used widely for mobile communications especially wireless modems and mobile phones.	
Typically used radio waves that provide a high-speed signal transmission over a long distance.	
Applications such as air navigation, television and radio broadcasting, weather forecasting, video conferencing, global positioning systems and internet connection use this type of transmission media.	

- (e) Identify the communication device based on the given function.

[6 marks]

Characteristic	Communication device
A device that allows computers and devices to transfer data wirelessly among themselves or to a wired network using wireless technologies.	
A central device that connects devices in a wired network but only sends data to the intended recipient.	
Used to connect two LANs.	

A communication device that enables a computer to communicate over analogue networking media.	
A device that enables a computer or device that does not have built-in networking capability to access a network.	
A device that connects multiple computers or other routers on a network.	

- 6 The web is a worldwide resource of information. People use internet technology to find and explore various information that has no limit.

(a) Define search tools.

[2 marks]

(b) Explain the characteristics of both types of search tools mentioned in 6(a).

[4 marks]

(c) Identify the type of search strategy below:

[4 marks]

Situation	Search Strategies
Use 'OR' to broaden your search by looking for several terms in a source whether they appear together or not.	
Use the special symbol "*" which allows you to search simultaneously for several words	
Using quotation marks (" ") makes it possible to search for important	

words/phrases instead of individual keyword	
Use particular search characteristics such as URL or website to limit the search result.	

- (d) Identify the types of web activities based on the given description following:

[4 marks]

Description	Types of Web Activity
Digital content is designed to engage and entertain individuals through media and services that can be accessed online.	
It involves online transactions, electronic payments and the exchange of products and services without the need for physical interactions.	
users can create and interact with content and other users on the internet.	
Prominent way for individuals to access up-to-date news content from a wide range of sources.	

- (e) Privacy issues are a fundamental aspect of data management in the digital area.

- i) Explain data privacy.

[2 marks]

- ii) Give **two (2)** examples of how to protect our privacy data while doing web activities.

[2 marks]

- (f) The goal of IoT is to enable any objects to communicate and interact with each other, as well as with humans to provide various services and capabilities.

Explain **one (1)** application of IoT in home appliances.

[2 marks]

END OF QUESTIONS.